
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 Conformal PID states long-run coverage for an arbitrary bounded scorecaster, and
2 a published corollary retains its rate under any predictable perturbation of the
3 deployed threshold inside an admissible radius. The opposite reading is in print too,
4 under overlapping authorship, and acted on. We make two claims. The record holds
5 both halves and we join them, conceding placement and derivation by name. And
6 at the null scorecaster the corollary’s condition is *tight*: a legal L_1 dead band on the
7 *completed* threshold leaves that set past $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, which
8 is the published radius itself there, and past it realised miscoverage is 1.000000
9 against $\alpha = 0.1$ while saturation fires every round. Leaving the set costs coverage
10 and not only the rate: partial adjustment there overruns Proposition 2’s bound
11 $42\times$ at $T = 10^6$ and still covers, 0.100035. The edge belongs to the saturator–
12 scorecaster pair, not to the readout, and elsewhere the condition is sufficient only:
13 under the equally legal $\hat{q} \equiv +b/2$ the failing band covers, and under $\hat{q} \equiv -b/2$
14 both the failing band and partial adjustment at $w = 0.999$ return 1.000000.

15 1 Introduction

16 An online conformal method emits a threshold before each observation and moves it afterwards. Fore-
17 cast stability names that movement, scores it and prices it [Tunc et al., 2013, Godahewa et al., 2025,
18 Van Belle et al., 2023, Pritularga and Kourtzes, 2024, Genov et al., 2026, Van Belle et al., 2026].
19 Varying a forecast’s temporal structure at fixed level was invented earlier, in forecast verification,
20 where the predictive marginal is matched on real producers [Gneiting et al., 2007, Pinson and Girard,
21 2012, Worsnop et al., 2018]; that coverage and mean length under-describe a deployed interval is
22 established too [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

23 **First: the record carries a claim and its refutation, and this paper joins them.** Conformal PID
24 [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator and asks of it
25 only that it be predictable and lie in $[-b/2, b/2]$ — “any function of the past” (Theorem 1) — so a
26 readout placed in that slot is already quantified over. Also in print, of the same method: “[s]ince
27 using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee” [Li and Rodríguez,
28 2024] — stated and acted on in that paper’s baselines paragraph, refereed and verbatim on printed
29 page 18443. The refutation is in print too, twice and with proof: Li et al. [2025, Corollary 2] add an
30 arbitrary predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, observe that the result
31 “has exactly the CPID error-integration form”, and conclude $|E_T| \leq ch(T) + 1 = o(T)$; the same
32 generic-slot argument carries AcMCP’s long-run coverage [Wang and Hyndman, 2024]. The first
33 and last authors of Li et al. [2025] are the authors of Li and Rodríguez [2024] and cite it: the halves
34 already share a bibliography, this is ordinary self-correction, and the joining is what this paper adds.
35 Four surfaces, queried on 20 August 2026 for the two papers’ identifiers and for “scorecaster” —
36 arXiv full text, arXiv abstracts, OpenReview, and the six works Semantic Scholar records as citing Li

37 and Rodríguez [2024] — return no document containing both the claim and the corollary. Hu et al.
 38 [2026] still calls scorecaster choice “arbitrary” and lacking “principled guidance”, while asserting
 39 valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

40 **Second: that admissible set has an edge, and the condition there is tight.** Corollary 2 is a
 41 *retention* result, so what it leaves open is outside that set. Section 3 exhibits a legal perturbation that
 42 leaves it — an L_1 dead band on the *completed* threshold, the family characterised here — and locates
 43 the edge at $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, the corollary’s own radius at the null scorecaster. Past
 44 it the price is coverage and not only the rate, so the published sufficient condition is necessary there.
 45 Those two claims are the paper; Section 4 sets them beside four vocabularies for the same movement.

46 2 Setup

47 **The producer.** A forecaster fit on data before t induces a score s_t , a threshold q_t is emitted before
 48 y_t is revealed, the deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$; the
 49 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
 50 [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

51 their iteration (5), in which r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of x once
 52 $|x| \geq c h(t)$, for h nonnegative, nondecreasing and sublinear. Writing $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$,
 53 Proposition 2 gives $|E_T| \leq c h(T) + 1$; boundedness is read on realised scores.

54 **One scalar, two readouts, two places to put it.** Damping the movement of a sequentially updated
 55 scalar has two standard readouts, both prior work: a quadratic cost gives linear partial adjustment
 56 towards the target, $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}_{t+1}^{\text{raw}} + \lambda\tilde{q}_t$ [Gârleanu and Pedersen, 2013], and a proportional
 57 cost a no-trade band, moving only when the target is more than τ away and then pulling back by
 58 exactly τ [Constantinides, 1986, Davis and Norman, 1990]. The first is published as post-processing
 59 [Godaheewa et al., 2025] and applied to a deployed conformal threshold [Binny and Dixit, 2025,
 60 Eq. 13]. It has two positions in (1): **(B)** in the $\hat{q}_{t+1}$ slot, where both readouts keep $\{\hat{q}_t\}$ in the convex
 61 hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set; and **(A)** downstream of
 62 the completed q_{t+1} , outside it. Placement A is measured here. *The dead band (L1) is not a stylistic*
 63 *alternative to the partial-adjustment (L2) form*; it is the specific perturbation family whose failure §3
 64 characterises. Neither is safe by construction: L2 forfeits Proposition 2’s rate while covering, L1 past
 65 τ^* loses coverage, and at the legal $\hat{q}_t \equiv -b/2$ both lose it. What is tight is the radius, not a form.

66 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
 67 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new: $\sum |\Delta q|$ over a
 68 predictive distribution is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it
 69 *jumpiness*, a *convergence index* and *(in)consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger
 70 et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length.
 71 Table 2’s caption names ten across four vocabularies, which is why a fresh one is used here.

72 **Harness.** $\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t + 2)$, and the clipped saturator $r_t(x) =$
 73 $b \text{clip}(x/(c h(t)), -1, 1)$, which meets condition (4) with equality. The scorecaster is null and
 74 legal, reducing (1) to the pure error integration Proposition 2 is stated about. Neither free choice is
 75 neutral: condition (4) is a lower bound this saturator meets exactly, and Theorem 1 permits score-
 76 casters $b/2$ either side of the null one. The adversary is specified rather than tie-broken: it plays
 77 $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$ — the $\varepsilon \downarrow 0$ reading of “sets the score
 78 at the threshold”, firing the strict indicator. Legality is Theorem 1’s $[-b/2, b/2]$ on both sides, so
 79 $b/2 + b/2 = b$ is exactly tight.

80 3 The edge of the admissible set, measured

81 **The retention result is published, and this section tests it from outside.**
 82 Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ on the de-
 83 ployed threshold keeps (1) in error-integration form and *retains* $|E_T| \leq c h(T) + 1$; the admissible

Table 1: Placement A: a one-scalar readout on the *completed* threshold; one deterministic adversary, one pass to $T = 10^6$. *All eleven arms run are listed*, so no covering arm is withheld. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, “Sat.” the rounds on which (4) delivers its full $\pm b$.

Readout on the completed threshold	miscov. 2×10^5	$\max_t E_t 10^6$	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	53.00	3.58	0.3181	16
running mean	0.099940	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	900,000	60,747	1.0000	0.5

radius is theirs. A downstream partial adjustment of weight w leaves it. Its excursion at $T = 10^4$ is 6.30/63.00/623.70 for $w = 0.9/0.99/0.999$, and for the last two it is *flat in T* while the bound grows like $\log T$, so the ratio falls with the horizon, 61.08 to 42.10; the measured law across the arms is $\max_t |E_t| \approx \max\{0.5 h(T) + 0.9, 0.63/(1 - w)\}$, and five of the eleven arms exceed the bound at $T = 10^6$. The forfeit is therefore a property of the smoothing *strength* and not of Placement A as such, and it buys something: the $w = 0.999$ arm cuts deployed travel from 28,311 to 202. The sign is the opposite of the intuitive one. A readout on the completed q_{t+1} touches neither r_t nor the integrator’s argument, so it cannot put condition (4) at risk; it delays the correction the integrator has already computed, so the accumulator excurses *further*. Smoothing the output does not damp E_t , it feeds it. The 623.70 is the primary regime’s figure — the adversary landing $\varepsilon = 10^{-9}$ above the deployed threshold under the harness’s strict indicator — and what it turns on is that ε rather than the counting convention: an adversary playing the threshold *exactly* returns 70.80 under the same indicator, because its signature move then covers.

The L_1 dead band leaves the radius, and the edge is measurable. A dead band pulls the deployed threshold back by exactly τ when it releases, so “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator’s ceiling: its realised supremum is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. The adversary’s best legal play is $b/2$, so coverage is lost exactly when that supremum drops below it, at $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. Every dead-band dead-band configuration in the variation sweep agrees, 19 of 19: three widths under each of the null scorecaster, $\hat{q} \equiv +b/2$, $\hat{q} \equiv -b/2$, a saturator at level $4b$ and Angelopoulos et al.’s tangent integrator, plus four wider bands at the null scorecaster. An eleven-point grid at the null scorecaster *locates* that edge between $\tau = 1.000$ and 1.001 ; the other four settings are bracketed at three widths and not located. **Here $\sup r_t = b$ and $\hat{q} \equiv 0$, so $\tau^* = b/2$** — one point of the admissible class, and both terms move it. At $\hat{q} \equiv +b/2$, legal, the same $\tau = 1.5$ band covers at $T = 10^6$ (0.100010, $\max_t |E_t| = 11.20$, inside the bound); at $\hat{q} \equiv -b/2$, equally legal, $\tau^* = 0$ and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$. Raising the saturator’s level to $4b$, admissible since (4) is a lower bound, raises $\sup_x r_t(x)$ and τ^* with it and cuts 623.70 to 120.60; under the unbounded tangent integrator Angelopoulos et al. [2023] report using in all their experiments, $\sup_x r_t(x)$ is unbounded and so is τ^* , the three widths run there all cover, $\tau = 1.5$ at 0.100001, and 623.70 falls to 20.10. **The method’s own integrator therefore sits inside the set at every width run here.** Outside it the transition is a step: $\tau = 1.000$ covers at $T = 10^5$ while $\tau = 1.001$ returns 1.000000, and past τ^* the threshold sits permanently below the reachable score range and $|E_t| = (1 - \alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; $\tau = 2.0, 2.5, 3.0$ and 5.0 fail too, so the failing set is (τ^*, ∞) , open at the left. **The endpoint is the one row in the record where the adversary’s convention decides the answer:** at $\tau = \tau^*$ the deployed supremum equals the adversary’s own cap $b/2$, so the primary regime covers while the exact-threshold play returns 1.000000 or 0.000000 according to the tie rule. Every other row is convention-free. **And the failing arms have Sat. = 1.0000:** condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2’s other hypothesis, that the threshold closing the loop is the integrator’s output.

What that measures is that the published condition is tight. At $\mu_t = 1$, $\hat{q}_t \equiv 0$ the measured τ^* is exactly Corollary 2’s radius, and crossing it costs coverage rather than the rate. **The claim is**

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness, convergence index, (in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	Forecast stability	Verification	Switching-cost OL
a distribution-free coverage guarantee for any bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	long-run- the scored movement functional, the readout map as post-processing [Godaheva et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2], [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control <i>stronger</i> than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013]

125 **that necessity: a tightness result about someone else’s sufficient condition**, smaller and more
126 precise than a forfeit of the rate by post-processing. Elsewhere the condition stays sufficient only
127 — at $\hat{q}_t \equiv +b/2$ Corollary 2 permits radius 0 while every width run there covers — which is what
128 makes “tight” a located claim.

129 **The constructive reading, and it is the corollary’s rather than ours.** Placement B keeps $\{\hat{q}_t\}$
130 inside the admissible radius, so Theorem 1 carries it already; and at Placement A, condition (4)
131 bounds $\sup_x r_t(x)$ only from below, so raising the saturator’s level widens the band at no cost to the
132 theorem. What the measurement adds is that the band has an edge, that the edge is computable before
133 deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is not a graceful degradation.

134 4 Where this sits

135 Four vocabularies name one quantity many ways, and the object that settles its validity — the
136 admissible set around the deployed threshold — sits in the first of them.

137 **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023] state
138 coverage for any admissible integrator and any scorecaster and run a smoothing-family model in
139 that slot themselves; Dupuy et al. [2026] give the generic argument in their Appendix A and Duerst
140 et al. [2024] adopt that scorecaster. The derivation is prior work too, and that is the concession that
141 matters most: Li et al.’s Corollary 2 proves the retention statement in Conformal PID’s own notation
142 [Li et al., 2025], with Wang and Hyndman [2024] an independent second instance. The smoother
143 object, the metric arithmetic and both readout maps are conceded in Section 2, and Zheng et al.
144 [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their
145 own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path,
146 where free-until-the-bound-binds is the defining specification rather than a finding [Galeani et al.,
147 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model*
148 update on the conformal score; their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$
149 for a sublinear h , supported with a plot rather than derived. That accumulation is what Section 3
150 measures diverging.

151 5 Limitations

152 **One construction, one adversary, one saturator**, and $0.63/(1-w)$ is specific to the saturator’s
153 *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it**
154 **applies.** Under the tangent integrator $h(t) = t/\log t$, so the certified gap $(ch(T) + 1)/T$ decays as
155 $O(1/\log T)$ rather than $O(1/T)$: what is forfeited was not a tight certificate. **The dead-band widths**
156 **are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$,
157 so the failing $\tau = 1.5$ is $0.75b$; under i.i.d. scores it returns 0.249376, which is not a retreat but an
158 adversary-free confirmation of the mechanism — the threshold pins at $b - \tau = 0.5$ and uniform
159 scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers**
160 **already read**, which carries less weight than a rerun done without sight of them. **Placement B is**
161 **conceded, not tested. One seed, and only the null scorecaster’s edge is located**; the other four
162 settings are bracketed at three widths. **What would overturn the claim** is a legal $(r_t, \hat{q}_t)$ pair with
163 $\tau \leq \tau^*$ that loses coverage, or $\tau > \tau^*$ that keeps it; the sweep has neither.

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